This model follows the **industry-oriented template** already defined and demonstrates how students should present **process flow, simulation steps, structure evolution, and analysis** using **SEMulator3D**.

This report is intended only as a **reference format**, not for direct copying.

SHALLOW TRENCH ISOLATION (STI) FORMATION SIMULATION USING SEMULATOR3D

Abstract

Shallow Trench Isolation (STI) is a critical isolation technique used in modern CMOS fabrication to electrically isolate adjacent transistors and reduce leakage currents. This project presents the simulation of STI formation using SEMulator3D process simulation software. The process flow includes trench lithography, anisotropic etching, oxide deposition, and chemical mechanical polishing (CMP). The evolution of the structure at each fabrication step is analyzed in a three-dimensional environment. Process parameters such as trench depth and oxide thickness are varied to study their impact on isolation quality and planarization. The results demonstrate how process simulation helps visualize fabrication challenges and optimize isolation structures before actual manufacturing. The project provides practical understanding of semiconductor process integration and isolation techniques used in industry.

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1. Introduction

1.1 Background

As CMOS technology scales to smaller nodes, device isolation becomes increasingly important to prevent leakage currents and latch-up effects. Earlier LOCOS isolation techniques suffered from bird’s beak effects and poor scalability. STI technology replaced LOCOS due to better scalability and planar surface formation.

1.2 Industrial Relevance

STI is widely used in:

- CMOS logic technologies
- FinFET fabrication
- Advanced memory devices
- Sub-100 nm technology nodes

STI ensures reduced parasitic capacitance and improved device packing density.

1.3 Project Objectives

- To simulate STI fabrication using SEMulator3D.
- To observe structure evolution during process steps.
- To analyze the effect of trench depth and oxide fill.
- To understand CMP planarization effects.

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2. Process Overview

2.1 Technology Node

180 nm CMOS Technology (Educational Reference Node)

2.2 Process Flow Summary

1. Silicon substrate preparation
2. Pad oxide formation
3. Nitride mask deposition
4. Lithography patterning
5. Trench etching
6. Oxide deposition (Trench fill)
7. Chemical Mechanical Polishing (CMP)
8. Nitride removal

2.3 Initial Structure Description

The starting structure consists of a silicon substrate with uniform thickness. A thin pad oxide layer is deposited to reduce stress between silicon and nitride layers.

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3. Simulation Methodology

3.1 Simulation Tool Description

SEMulator3D enables visualization of process steps in a three-dimensional environment and allows users to simulate deposition, etching, and planarization processes with realistic geometrical evolution.

3.2 Process Steps Implemented

Step No	Process Step	Purpose
1	Pad Oxide Deposition	Stress relief layer

Step No	Process Step	Purpose
2	Nitride Deposition	Hard mask formation
3	Lithography	Trench pattern definition
4	Anisotropic Etch	Trench creation
5	Oxide Deposition	Trench filling
6	CMP	Surface planarization

3.3 Process Parameters Used

Parameter	Value	Unit	Reason
Trench depth	300	nm	Isolation requirement
Oxide thickness	400	nm	Complete trench fill
Nitride thickness	150	nm	Etch resistance

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4. Structure Development

4.1 Intermediate Structure Evolution

Step 1: Pad Oxide Formation
A thin oxide layer is deposited on silicon to reduce mechanical stress.

Step 2: Nitride Mask Deposition
A silicon nitride layer is deposited to act as an etch mask.

Step 3: Lithography and Patterning
Photoresist defines trench regions.

Step 4: Trench Etching
Anisotropic etching creates vertical trenches in silicon.

Step 5: Oxide Fill
Oxide deposition fills the trench completely.

Step 6: CMP Planarization
Excess oxide is removed to achieve a flat surface.

(Students should insert SEMulator3D screenshots here after each step.)

4.2 Final Structure

The final structure shows oxide-filled trenches with planar surface suitable for subsequent gate formation steps.

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5. Results and Analysis

5.1 Geometry Analysis

- Trench depth achieved: 300 nm
- Sidewall angle: approximately 90°
- Uniform oxide fill observed

5.2 Process Sensitivity Study

Increasing trench depth beyond 350 nm resulted in incomplete oxide fill before CMP, leading to void formation.

5.3 Observations

- CMP is critical for surface planarity.
- Excess oxide deposition increases CMP time.
- Proper trench depth ensures effective isolation.

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6. Discussion

The simulation demonstrates the importance of balancing etch depth and deposition thickness. Improper process tuning may lead to dishing or erosion during CMP, affecting device performance.

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7. Industrial Insights

In real semiconductor fabrication, STI defects can lead to leakage and yield loss. Process simulation tools like SEMulator3D help engineers optimize process conditions before production, reducing manufacturing cost and cycle time.

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8. Conclusion

The STI formation process was successfully simulated using SEMulator3D. The project provided insight into process integration, trench formation, and planarization challenges encountered in modern CMOS fabrication.

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9. Future Scope

- STI scaling for advanced nodes
- Integration with FinFET structures
- Stress analysis studies
- Process variation analysis

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10. References

[1] S. Wolf, *Silicon Processing for the VLSI Era*, Lattice Press.
[2] Plummer, Deal & Griffin, *Silicon VLSI Technology*, Pearson.
[3] Lam Research Process Documentation.

Guideline for Students

- Replace parameter values with your own simulation data.
- Insert structure images from SEMulator3D.
- Provide analysis, not only screenshots.
- Explain process decisions like an engineer.
